

Asynchronous Multi-Agent GRPO + OPD

Four Training Settings and Complete Workflow

Method record | October 3, 2026

Abstract

This note records Terminal and Tool-call trajectory organization with PRM disabled or enabled. All four settings share independent rollouts, asynchronous snapshot broadcasts, receiver-local training, and outcome-calibrated top-4 peer OPD. The confirmed baseline disables PRM. The enabled settings describe a structural extension with unresolved weights and boundary rules kept as explicit interfaces. No experimental conclusions or literature references are included.

1 Four settings and notation

Setting	PRM	Storage	GRPO advantage
Terminal	Off	One sample per step	One A_i^G per trajectory
Terminal	On	One sample per step	Step-specific $A_{i,t}^G$
Tool-call	Off	One long trajectory sample	One A_i^G for all action tokens
Tool-call	On	Long sample with step spans	$A_{i,t}^G$ assigned by span

Each batch contains 16 tasks and eight complete trajectories per task. Index q denotes a task, j its candidate trajectory, and i a complete trajectory in the batch. Index t denotes an interaction step, k an output token within that step, and T_i the number of steps.

$$Q = 16, \quad G = 8, \quad B = QG = 128, \quad i = (q - 1)G + j. \quad (1)$$

An action $a_{i,t}$ contains generated reasoning text, a tool call, or a final answer. Execution produces observation $o_{i,t}$. The input for step t uses only observations already received and retained history:

$$\begin{aligned} s_{i,t} &= \text{Format}(\text{system}, x_q, o_{i,0}, a_{i,1}, o_{i,1}, \dots, a_{i,t-1}, o_{i,t-1}), \\ h_{i,t,k} &= (s_{i,t}, y_{i,t,<k}), \quad a_{i,t} = (y_{i,t,1}, \dots, y_{i,t,L_{i,t}}). \end{aligned} \quad (2)$$

The formatted input $s_{i,t}$ is not the full environment state. Prefix $h_{i,t,k}$ is the actual context before the token. History retention and truncation follow the input interface. With PRM off, tool observations do not automatically become numerical process rewards.

2 Shared architecture and independent learning

The system contains a task pool, a strong external outcome judge, and N independent agents. Each agent owns environment instances, trajectories, trainable parameters, and an inbox. All effective policies start identically:

$$\theta_a^{(0)} = \theta_0, \quad a \in \{1, \dots, N\}. \quad (3)$$

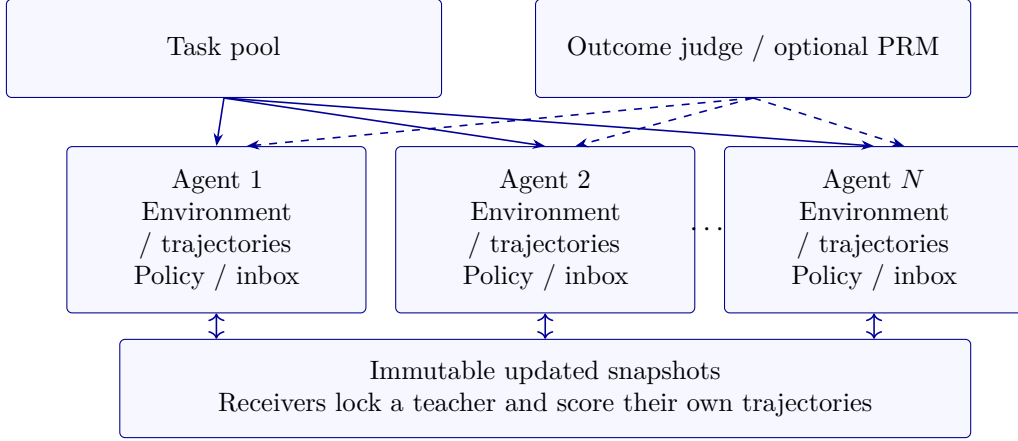


Figure 1: Shared interfaces and independent learners. Optional PRM scoring does not change the role of peer snapshots.

2.1 Four policy roles

- π_{old} : the frozen local behavior policy for all 128 trajectories and the denominator of policy ratios.
- π_{θ} : the current local policy, initialized from the old model. Gradients update only its designated trainable parameters.
- π_T : a frozen peer teacher selected from this round’s inbox to score the receiver’s own prefixes.
- π_{ref} : the reference policy for KL regularization, with a role distinct from the peer teacher.

Rollouts remain locally on-policy: broadcasts do not replace the behavior model or supply other agents’ trajectories. Every completed local update broadcasts a new snapshot, including joint GRPO + OPD updates. No agent waits for others to finish the same round.

3 Inbox, teacher locking, and asynchronous timing

3.1 Round-local eligibility window

Index a denotes an agent and b its local batch. At new-batch intake, clear the entire inbox, including snapshots received during the previous scoring or update phase. Let $t_{a,b}^{\text{start}}$ be the batch start and $t_{a,e}^{\text{arr}}$ the receiver-side arrival time of entry e :

$$\mathcal{C}_{a,b} = \{e : t_{a,b}^{\text{start}} \leq t_{a,e}^{\text{arr}} \leq t_{a,b}^{\text{A}}\}. \quad (4)$$

Event $t_{a,b}^{\text{A}}$ means that all required rewards and GRPO advantages are ready. With PRM on, this includes the required process scores. Entries identify the sender, its local update number, an immutable snapshot, and its arrival time. An empty inbox yields local GRPO plus reference KL. Otherwise, select the latest arrival at advantage completion; select uniformly only among entries tied for the latest time:

$$\begin{aligned} u_{a,b} &= \max_{e \in \mathcal{C}_{a,b}} t_{a,e}^{\text{arr}}, \\ \mathcal{E}_{a,b} &= \{e \in \mathcal{C}_{a,b} : t_{a,e}^{\text{arr}} = u_{a,b}\}, \\ e_{a,b}^* &\sim \text{Unif}(\mathcal{E}_{a,b}), \quad \pi_T = \pi_{e_{a,b}^*}^{\text{snapshot}}. \end{aligned} \quad (5)$$

An arrival exactly at advantage completion is eligible. Lock the teacher through optimization. Later broadcasts enter the inbox but do not change the teacher, scores, or branch; discard them at the next batch start. Latest arrival does not imply best performance, and sender-local update numbers are not global versions.

3.2 Events and coordination

$$t_{a,b}^{\text{roll}} \leq t_{a,b}^{\text{A}} = t_{a,b}^{\text{select}} \leq t_{a,b}^{\text{T,start}} \leq t_{a,b}^{\text{T}} \leq t_{a,b}^{\text{opt,start}}. \quad (6)$$

Timestamp	Event for the same agent and batch
$t_{a,b}^{\text{roll}}$	All 128 complete rollouts finish
$t_{a,b}^{\text{A}}$	All required rewards and GRPO advantages are ready
$t_{a,b}^{\text{select}}$	Logical teacher selection and locking
$t_{a,b}^{\text{T,start}}$	The locked teacher starts scoring this batch
$t_{a,b}^{\text{T}}$	All required teacher scores are ready
$t_{a,b}^{\text{opt,start}}$	Local parameter optimization begins

Skip teacher scoring if no teacher exists. PRM may score during interaction; the finally selected peer starts batch scoring after advantage completion. Different agents overlap asynchronously. A local update pauses intake while other agents continue. If all are updating, intake pauses. Shutdown finishes in-flight rollouts, scoring, and updates, and broadcasts completed updates.

4 PRM off: trajectory-level GRPO advantages

4.1 Outcome reward and task groups

Let z_i be the final answer and $\mathcal{V}(x_q, z_i)$ the external judge’s correctness label. A judged final answer receives the confirmed outcome reward:

$$O_i = \begin{cases} +1, & \mathcal{V}(x_q, z_i) = 1, \\ -1, & \mathcal{V}(x_q, z_i) = 0. \end{cases} \quad (7)$$

Intermediate execution results, errors, and test output enter later context without automatically becoming numerical process scores. A missing terminal reward must not silently become -1 ; failure handling remains an unspecified input interface. A GRPO group contains eight complete trajectories from the same initial task. Count each trajectory’s reward once:

$$\mu_q = \frac{1}{8} \sum_{j=1}^8 O_{q,j}, \quad \sigma_q = \sqrt{\frac{1}{7} \sum_{j=1}^8 (O_{q,j} - \mu_q)^2}. \quad (8)$$

Use a sample standard deviation with denominator seven, replacing it by a positive small quantity only when it is zero:

$$\tilde{\sigma}_q = \begin{cases} \sigma_q, & \sigma_q > 0, \\ \epsilon_{\text{std}}, & \sigma_q = 0, \end{cases} \quad A_{q,j}^G = \frac{O_{q,j} - \mu_q}{\tilde{\sigma}_q}, \quad \epsilon_{\text{std}} > 0. \quad (9)$$

4.2 Broadcasting advantages and update timing

Let $M_{i,t}^X$ denote effective output-token positions in step t for branch X , which can be GRPO, peer OPD, or reference KL. Copy the trajectory advantage unchanged to its action tokens:

$$\boxed{A_{i,t,k}^G = A_i^G}, \quad k \in M_{i,t}^G. \quad (10)$$

Do not divide by the number of steps or tokens. Different probability ratios still produce different token losses. Optimize after collecting and aggregating the batch, rather than running a separate GRPO update after every transition. For example, if a three-step trajectory has $A_i^G = 0.8$, all effective output tokens in all three steps use 0.8, including an early failed attempt. This signal describes the complete trajectory’s relative outcome, not the correctness of every action. If all eight candidates succeed or all fail, both the numerator and standard deviation are zero, yielding zero GRPO advantages. Retain the group; peer OPD and reference KL may still contribute. These rules are identical for Terminal and Tool-call with PRM off.

5 PRM on: step-level GRPO advantages

5.1 Compose rewards before step normalization

Score $P_{i,t}$ evaluates whether the current reasoning and action advance the task. The task, actual history, action, and resulting observation can inform this judgment; an execution error does not necessarily imply an unhelpful action. Symbol η denotes the unresolved process-reward weight:

$$R_{i,t} = O_i + \eta P_{i,t}. \quad (11)$$

The same outcome O_i contributes to every step, while process scores vary. Compose $R_{i,t}$ first, then compare within a group. Do not normalize outcomes and PRM scores separately and add their advantages. Groups use the same task and interaction-step index. Because lengths vary, define the candidates that actually contain the step:

$$\mathcal{I}_{q,t} = \{j \in \{1, \dots, 8\} : T_{q,j} \geq t\}, \quad n_{q,t} = |\mathcal{I}_{q,t}|. \quad (12)$$

First consider eight candidates that all contain this step and have valid scores, retaining the sample-standard-deviation convention:

$$\mu_{q,t} = \frac{1}{8} \sum_{j=1}^8 R_{q,j,t}, \quad \sigma_{q,t} = \sqrt{\frac{1}{7} \sum_{j=1}^8 (R_{q,j,t} - \mu_{q,t})^2}, \quad n_{q,t} = 8. \quad (13)$$

$$\tilde{\sigma}_{q,t} = \begin{cases} \sigma_{q,t}, & \sigma_{q,t} > 0, \\ \epsilon_{\text{std}}, & \sigma_{q,t} = 0, \end{cases} \quad (14)$$

$$\boxed{A_{q,j,t,k}^G = A_{q,j,t}^G} = \frac{R_{q,j,t} - \mu_{q,t}}{\tilde{\sigma}_{q,t}}.$$

Here $n_{q,t} = 8$ and the denominator is seven. Every effective output token in a step shares $A_{i,t}^G$ without division by token count.

5.2 Interpretation and boundaries

A successful trajectory can contain a negative step advantage. Equal terminal rewards can still yield nonzero advantages when process scores differ. A positive PRM score does not guarantee a positive relative advantage; the composed reward’s group-relative value matters. For $n_{q,t} < 8$, whether to normalize the existing-step subgroup remains unresolved, as does handling missing or failed PRM scores. This note does not pad absent steps, fill zeros, drop samples, or fix η . PRM scoring may run asynchronously, but an update waits for the required outcomes, process scores, and advantages of its batch.

6 Shared mechanism: outcome-calibrated peer signals

6.1 Teacher preference for the actual final answer

After locking the teacher, score the receiver’s own trajectory with teacher forcing. Set F_i contains actual final-answer token positions (t, k) . Its extraction boundary remains an input interface; the full sequence of tool feedback is not automatically the final answer.

$$J_i = \frac{1}{|F_i|} \sum_{(t,k) \in F_i} [\log \pi_T(y_{i,t,k} \mid h_{i,t,k}) - \log \pi_{\text{old}}(y_{i,t,k} \mid h_{i,t,k})], \quad |F_i| > 0. \quad (15)$$

Quantity J_i averages teacher-minus-old log probabilities on the same actual answer. Use full-vocabulary probabilities without clipping this difference first. A positive value means greater teacher support for the answer; a negative value means less support. It is not the teacher’s own solving accuracy.

6.2 Keep, reverse, or disable peer OPD

$$\lambda_i = \begin{cases} 0, & |F_i| = 0, \\ +1, & |F_i| > 0 \wedge (O_i J_i > 0 \vee J_i = 0), \\ -1, & |F_i| > 0 \wedge O_i J_i < 0. \end{cases} \quad (16)$$

O_i	J_i	λ_i	Signal direction
+1	> 0	+1	Keep: teacher favors a correct answer
+1	< 0	−1	Reverse: teacher disfavors a correct answer
−1	< 0	+1	Keep: teacher disfavors an incorrect answer
−1	> 0	−1	Reverse: teacher favors an incorrect answer

For a nonempty answer with $J_i = 0$, set $\lambda_i = +1$: a zero mean difference need not imply zero local differences. For an empty answer, set $\lambda_i = 0$, do not compute J_i , and explicitly set peer OPD to zero while retaining GRPO groups and advantages. An empty answer is distinct from an empty action-token set. With no teacher, omit batch peer OPD and retain GRPO plus reference KL. With a teacher and an empty answer, the zero OPD term still contributes to the batch denominator of 128; do not renormalize over nonzero terms. With PRM enabled, O_i remains the outcome reward and λ_i remains trajectory-level. Do not automatically substitute $R_{i,t}$ or create step-specific $\lambda_{i,t}$. PRM changes GRPO advantages, not this direction rule.

7 Shared mechanism: old-policy top-4 peer OPD

7.1 Frozen candidates and probability weights

At each effective output prefix $h_{i,t,k}$, select the local old policy’s top four tokens. Freeze candidate identities throughout scoring and optimization. The teacher and current policy score these same candidates:

$$S_{i,t,k} = \text{Top4}(\pi_{\text{old}}(\cdot \mid h_{i,t,k})), \quad |S_{i,t,k}| = 4. \quad (17)$$

$$\ell_{i,t,k,v}^X = \log \pi_X(v \mid h_{i,t,k}), \quad X \in \{\text{old}, T, \theta\}. \quad (18)$$

All probabilities come from the full-vocabulary softmax. Only candidate weights w are normalized within the top-four set:

$$w_{i,t,k,v} = \frac{\exp(\ell_{i,t,k,v}^{\text{old}})}{\sum_{u \in S_{i,t,k}} \exp(\ell_{i,t,k,u}^{\text{old}})}, \quad \sum_{v \in S_{i,t,k}} w_{i,t,k,v} = 1. \quad (19)$$

7.2 Difference clipping, direction, and gradients

$$d_{i,t,k,v} = \ell_{i,t,k,v}^T - \ell_{i,t,k,v}^{\text{old}}, \quad \Delta_{i,t,k,v} = \text{clip}(d_{i,t,k,v}, -C, C). \quad (20)$$

$$\boxed{a_{i,t,k,v}^{\text{peer}} = \text{sg}[\lambda_i w_{i,t,k,v} \Delta_{i,t,k,v}]}, \quad \lambda_i \in \{-1, +1\}. \quad (21)$$

Threshold C remains symbolic. Operator sg stops gradients. The old policy, teacher, candidate identities, weights, differences, and λ_i are fixed. Gradients flow through current-policy probabilities to local trainable parameters. For $\lambda_i = 0$, use the explicit zero OPD branch. Each output position has four candidate advantages, not a PRM scalar simply added to the sampled token. Do not divide their loss sum by four: w already normalizes the candidates. The sampled token may fall outside the top four, so separately retain actual answer-token log probabilities needed for J_i . Score the teacher, old policy, and current policy under the same actual prefix, retaining tool feedback as presented during rollout. The teacher does not execute an alternative environment path. For $\lambda_i = -1$, reverse candidate advantages before min/clip ; negating the ordinary clipped OPD loss afterward is a different objective.

8 Separate clipping, two-level averaging, and updates

GRPO uses the sampled-token local policy ratio; OPD uses ratios for top-four candidates. Neither denominator is the teacher:

$$\begin{aligned} r_{i,t,k} &= \frac{\pi_\theta(y_{i,t,k} \mid h_{i,t,k})}{\pi_{\text{old}}(y_{i,t,k} \mid h_{i,t,k})}, \\ \rho_{i,t,k,v} &= \frac{\pi_\theta(v \mid h_{i,t,k})}{\pi_{\text{old}}(v \mid h_{i,t,k})}. \end{aligned} \quad (22)$$

$$c_X(r) = \text{clip}(r, 1 - \epsilon_X^{\text{low}}, 1 + \epsilon_X^{\text{high}}), \quad X \in \{\text{G}, \text{D}\}. \quad (23)$$

Apply PPO clipped surrogates independently. The PRM switch determines A^{G} :

$$L_{i,t,k}^{\text{G}} = -\min\{r_{i,t,k} A_{i,t,k}^{\text{G}}, c_{\text{G}}(r_{i,t,k}) A_{i,t,k}^{\text{G}}\}. \quad (24)$$

$$\begin{aligned} U_{i,t,k}^{\text{peer}} &= -\sum_{v \in S_{i,t,k}} \min\left\{\rho_{i,t,k,v} a_{i,t,k,v}^{\text{peer}}, \right. \\ &\quad \left. c_{\text{D}}(\rho_{i,t,k,v}) a_{i,t,k,v}^{\text{peer}}\right\}, \end{aligned} \quad (25)$$

$$L_{i,t,k}^{\text{peer}} = \begin{cases} 0, & \lambda_i = 0, \\ U_{i,t,k}^{\text{peer}}, & \lambda_i \in \{-1, +1\}. \end{cases}$$

8.1 Token means within trajectories, then trajectory means

For branch X , aggregate effective output tokens across all steps under its mask, then average complete trajectories:

$$\begin{aligned} \bar{L}_i^X &= \frac{\sum_{t=1}^{T_i} \sum_{k \in M_{i,t}^X} L_{i,t,k}^X}{\sum_{t=1}^{T_i} |M_{i,t}^X|}, \quad \sum_t |M_{i,t}^X| > 0, \\ \mathcal{L}_{\text{total}} &= \frac{1}{B} \sum_{i=1}^B \left(\alpha \bar{L}_i^{\text{G}} + \beta \bar{L}_i^{\text{peer}} \right) + 0.01 \mathcal{L}_{\text{KL}}, \quad B = 128. \end{aligned} \quad (26)$$

Each trajectory has batch weight $1/128$. Do not first average steps equally, average all expanded Terminal samples, or weight batch tokens equally. Keep α, β symbolic. Do not add the two advantages before a single clipping operation. Reference KL uses the same reduction. Symbol κ_θ denotes the selected token-level reference penalty, not another model:

$$\mathcal{L}_{\text{KL}} = \frac{1}{B} \sum_{i=1}^B \frac{\sum_t \sum_{k \in M_{i,t}^{\text{K}}} \kappa_\theta(h_{i,t,k}, y_{i,t,k}; \pi_{\text{ref}})}{\sum_t |M_{i,t}^{\text{K}}|}. \quad (27)$$

Reference identity, the exact KL implementation, and its mask remain interfaces; the coefficient is fixed at 0.01. Gradients flow to the current local policy without multiplication by λ_i . An empty action mask has an undefined denominator and unresolved handling, distinct from an empty final answer. Backpropagate this joint loss to optimize local parameters; no additional GRPO operation follows.

9 Settings 1 and 2: Terminal algorithms and workflows

Terminal splits a complete trajectory into step samples, retaining each actual input, current output, and complete-trajectory identity:

$$\tau_i \mapsto \{(s_{i,t}, a_{i,t}, i, t) : t = 1, \dots, T_i\}. \quad (28)$$

Sample	Current input	Current training target
Step 1	Task and initial observation $s_{i,1}$	Reasoning and action $a_{i,1}$
Step 2	$s_{i,2}$ retaining step 1 and feedback	Reasoning and action $a_{i,2}$
Step 3	$s_{i,3}$ retaining earlier interactions	Reasoning and answer $a_{i,3}$

Earlier actions are input history in a later sample, not repeated output targets in that sample. Teacher scoring uses exactly the same current input.

9.1 Setting 1: Terminal with PRM off

1. Freeze the old policy; collect eight complete trajectories per task for 16 tasks. Store step inputs, outputs, old log probabilities, and trajectory identities.
2. Obtain O_i from the external judge. Compute A_i^G among complete trajectories for each task, counting each reward once.
3. Assign A_i^G to all step-output tokens. At advantage completion, lock an inbox teacher and compute shared peer OPD signals.
4. Compute both step losses, recover each complete trajectory’s token mean, average 128 trajectories, add KL, optimize, and broadcast.

9.2 Setting 2: Terminal with PRM on

1. Retain rollout and step-sample organization; additionally obtain each $P_{i,t}$.
2. After O_i is ready, compose $R_{i,t} = O_i + \eta P_{i,t}$. Group by task and step to obtain $A_{i,t}^G$.
3. Assign $A_{i,t}^G$ to effective output tokens in step sample t . Lock a teacher when all required batch advantages are ready.
4. Retain OPD’s J_i, λ_i , and top-four rules. Average complete trajectories, optimize locally, and broadcast.

A trajectory split into seven samples still has batch weight 1/128. PRM changes GRPO supervision without changing outcome-based OPD gating or introducing per-step updates.

10 Settings 3 and 4: Tool-call algorithms and workflows

Tool-call stores accumulated model outputs and tool observations as one complete long sample. Each output position retains its actual context. Causal attention excludes future observations.

Sequence segment	Mask	PRM-off advantage	PRM-on advantage
Task and prompt	0	No direct action loss	No direct action loss
Step 1 reasoning/call	1	A_i^G	$A_{i,1}^G$
Tool observation 1	0	No direct action loss	No direct action loss
Step 2 reasoning/call	1	A_i^G	$A_{i,2}^G$
Tool observation 2	0	No direct action loss	No direct action loss
Final reasoning/answer	1	A_i^G	$A_{i,3}^G$

A zero loss mask does not hide tool feedback from later generation. Let p be a long-sequence position and ϕ_i map a step-local token to it:

$$p = \phi_i(t, k), \quad h_{i,p} = h_{i,t,k}, \quad A_{i,p}^G = A_{i,t,k}^G. \quad (29)$$

10.1 Setting 3: Tool-call with PRM off

1. Generate actions with the old policy, execute tools, and append actual observations. Record output masks and old log probabilities for 128 complete long trajectories.
2. Obtain outcomes O_i and compute A_i^G once among eight same-task trajectories. Assign it to all effective output positions of trajectory i .
3. At advantage completion, lock an inbox teacher. Under the same prefixes containing tool feedback, compute J_i, λ_i , and top-four OPD.
4. Compute action losses only at output positions. Average tokens within each trajectory, average trajectories, optimize, and broadcast.

10.2 Setting 4: Tool-call with PRM on

1. Retain the long sequence and mask; additionally record output spans and obtain $P_{i,t}$.
2. Compose $R_{i,t}$ after outcome scoring. Normalize by task and step and assign $A_{i,t}^G$ unchanged to its output span.
3. Lock a teacher at advantage completion. Retain outcome-calibrated OPD and top-four scoring at each output prefix.
4. Clip separately, average within trajectories and across the batch, add reference KL, optimize, and broadcast.

Long-sequence storage does not require a uniform advantage. With identical contexts, effective positions, and reduction, step samples and long sequences can express the same training objective.

11 Unified event loop and method boundaries

11.1 Complete algorithm for each agent

Run this loop independently. Terminal/Tool-call organization and the PRM switch determine sample representation and GRPO advantages.

1. **Start a batch.** Clear the old inbox. Freeze the old policy, initialize the current policy from local old parameters, and collect eight complete trajectories for each of 16 tasks.
2. **Interact and receive.** Generate reasoning and actions; execute them and build the next input from observations. Store broadcasts in this round’s inbox. Obtain step scores if PRM is on.
3. **Judge and build advantages.** Obtain all outcome rewards. With PRM off, normalize complete trajectories per task. With PRM on, compose step rewards before step grouping. Assign each advantage unchanged to its effective output tokens.
4. **Lock a teacher.** At advantage completion, include simultaneous arrivals and select the latest snapshot, uniformly breaking latest-time ties. An empty inbox yields GRPO + KL; otherwise lock the teacher.
5. **Score local trajectories.** Score actual prefixes with the frozen teacher. Retain actual-answer and old-policy top-four log probabilities; compute $J_i, \lambda_i, w, \Delta, a^{\text{peer}}$.
6. **Optimize locally.** Recompute current-policy ratios and separate GRPO/OPD clipped losses. Average tokens per trajectory, then 128 trajectories, and combine with KL. Update only designated local parameters.
7. **Broadcast and continue.** Immediately broadcast an immutable updated snapshot. If intake remains open, start another batch and clear the inbox again. Late arrivals do not change a locked teacher.
8. **Drain at shutdown.** Finish in-flight rollouts, scoring, and updates; broadcast completed updates. The stopping trigger remains an external input.

11.2 Confirmed conventions and extension boundaries

The confirmed baseline disables PRM. The enabled settings are structural extensions, not replacements. Keep η, α, β, C and clipping bounds symbolic. Do not invent defaults for absent steps, missing rewards, or empty action masks. Experimental, decoding, and compute-budget configurations are outside this note. Retain the user’s sample-standard-deviation, zero-variance replacement, and constant-group retention conventions. OpenClaw-RL’s current step path uses population standard deviation plus a small quantity and filters some constant groups; these behaviors are not adopted. Terminal composes rewards with a process coefficient. Tool-call’s step field directly uses $O_i + P_{i,t}$, while its coefficient affects another trajectory score. Symbol η describes a unified extension, not an identical implementation claim about both source branches. Peer calibration guarantees neither local action correctness nor monotonic accuracy. This record contains no training runs or experimental results.